

Flutter Exam

MaterialApp

```
import 'package:flutter/material.dart';

void main() {
  runApp(
    MaterialApp(
      debugShowCheckedModeBanner: false,
      home: HomePage(),
    ),
  );
}
```

Scaffold

```
Scaffold(
  appBar: AppBar(
    title: Text("Home"),
  ),
  body: Center(
    child: Text("Hello"),
  ),
);
```

AppBar

```
AppBar(  
  title: Text("My App"),  
  centerTitle: true,  
  backgroundColor: Colors.blue,  
  
  actions: [  
    Icon(Icons.search),  
  ],  
);
```

Body

```
body: Container(  
  padding: EdgeInsets.all(20),  
  
  child: Text(  
    "Body Content",  
  ),  
),
```

Text

```
Text(  
  "Hello Flutter",  
  
  style: TextStyle(  
    fontSize: 22,  
    fontWeight: FontWeight.bold,  
    color: Colors.black,  
  ),  
);
```

TextFormField

Basic Example

```
TextFormField(  
  decoration: InputDecoration(  
    labelText: "Username",  
    border: OutlineInputBorder(),  
  ),  
);
```

TextFormField with Prefix & Suffix Icons

```
TextFormField(  
  keyboardType: TextInputType.visiblePassword,  
  controller: password,  
  obscureText: true,  
  validator: (value)  
  {  
    if (value == null || value.isEmpty)  
    {  
      return "Password Can't be Empty";  
    }  
  }  
  
  decoration: InputDecoration(  
    labelText: "Password",  
    hintText: "Enter password",  
  
    prefixIcon: Icon(Icons.lock),  
  
    suffixIcon: Icon(Icons.visibility),  
  
    border: OutlineInputBorder(  
      borderRadius: BorderRadius.circular(12),  
    ),  
  ),  
);
```

SizedBox

Height Space

```
SizedBox(height: 20),
```

Width Space

```
SnackBar(width: 20),
```

SnackBar with Child

```
SnackBar(  
  width: 200,  
  height: 100,  
  
  child: Container(  
    color: Colors.blue,  
  ),  
);
```

ElevatedButton

```
ElevatedButton(  
  style: ElevatedButton.styleFrom(  
    backgroundColor: Colors.blue,  
  ),  
  
  onPressed: () {  
    print("Clicked");  
  },  
  
  child: Text("Login"),  
);
```

IconButton

```
IconButton(  
  onPressed: () {},  
  
  icon: Icon(Icons.favorite),  
);
```

TextButton

```
TextButton(  
  onPressed: () {},  
  
  child: Text("Register"),  
);
```

Row

```
Row(  
  mainAxisAlignment: MainAxisAlignment.spaceEvenly,  
  
  children: [  
    Icon(Icons.home),  
    Icon(Icons.favorite),  
    Icon(Icons.person),  
  ],  
);
```

Column

```
Column(  
  mainAxisAlignment: MainAxisAlignment.center,  
  crossAxisAlignment: CrossAxisAlignment.start,  
  
  children: [  
    Text("Name"),  
    Text("Email"),  
    Text("Phone"),  
  ],  
);
```

Navigation

Move to Another Screen

```
Navigator.push(  
  context,  
  
  MaterialPageRoute(  
    builder: (context) => SecondPage(),  
  ),  
);
```

Passing Data Between Screens

First Screen

```
ElevatedButton(  
  onPressed: () {  
  
    Navigator.push(  
      context,  
  
      MaterialPageRoute(  
        builder: (context) => SecondPage(  
          name: "Mohamed",  
        ),  
      ),  
    );  
  },  
  child: Text("Go"),  
);
```

Second Screen

```
class SecondPage extends StatelessWidget {  
  
  final String name;  
  
  SecondPage({required this.name});  
  
  @override  
  Widget build(BuildContext context) {  
  
    return Scaffold(  
  
      appBar: AppBar(  
        title: Text("Second Page"),  
      ),  
  
      body: Center(  
        child: Text(name),  
      ),  
    );  
  }  
}
```

List View

```
ListView(  
  children: [  
  
    ListTile(  
      leading: Icon(Icons.person),  
      title: Text("Mohamed"),  
      subtitle: Text("Flutter Developer"),  
    ),  
  
    ListTile(  
      leading: Icon(Icons.person),  
      title: Text("Ali"),  
      subtitle: Text("Mobile Developer"),  
    ),  
  
  ],  
);
```

ListView.builder

```
List<String> names = [
  "Mohamed",
  "Ali",
  "Ahmed",
  "Omar",
];

ListView.builder(
  itemCount: names.length,
  itemBuilder: (context, index) {
    return ListTile(
      leading: Icon(Icons.person),
      title: Text(
        names[index],
      ),
    );
  },
);
```

formKey variable

```
final _formKey = GlobalKey<FormState>();
```